Day 2 – AWS Global Infrastructure & Virtualization Fundamentals

Today's Topics

- **Methods of Connecting to AWS Cloud**
- **AWS Global Infrastructure**
- **Regions & Availability Zones**
- **Data Centers & Physical Servers**
- **High Availability (HA)**
- **Disaster Recovery (DR)**
- **Virtualization & Hypervisors**
- **Business Value of Cloud**

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1. Methods of Connecting to the Cloud

Cloud providers like Amazon Web Services provide multiple ways to connect and manage infrastructure.

A. AWS Console (Portal)

AWS Console is a web-based interface used through a browser.

Uses

- **Create servers**
- **Manage storage**
- **Configure networking**
- **Monitor applications**
- **Check billing**

Architecture Flow

User → Browser → AWS Console → AWS Services

Real-Time Example

Open:

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- **aws.amazon.com**
- **Login**
- **Open EC2 Service**
- **Launch Instance**

B. AWS CLI (Command Line Interface)

CLI allows engineers to manage AWS resources using commands.

Example Command

aws ec2 describe-instances

Benefits

- **Faster automation**
- **Useful for scripting**
- **Preferred in DevOps**

Diagram

User → Terminal → AWS CLI → AWS Infrastructure

C. Terraform / Infrastructure as Code (IaC)

Terraform creates infrastructure using code files.

Example

```
resource "aws_instance" "server" {
  ami         = "ami-12345"
  instance_type = "t2.micro"
}
```

Diagram

Terraform Code → Terraform Engine → AWS Resources

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2. AWS Global Infrastructure

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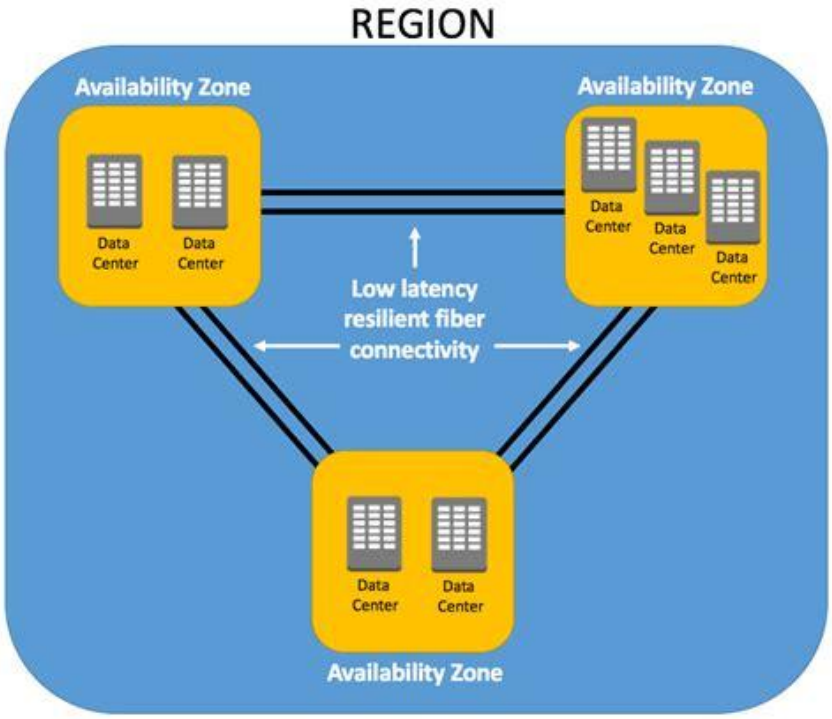
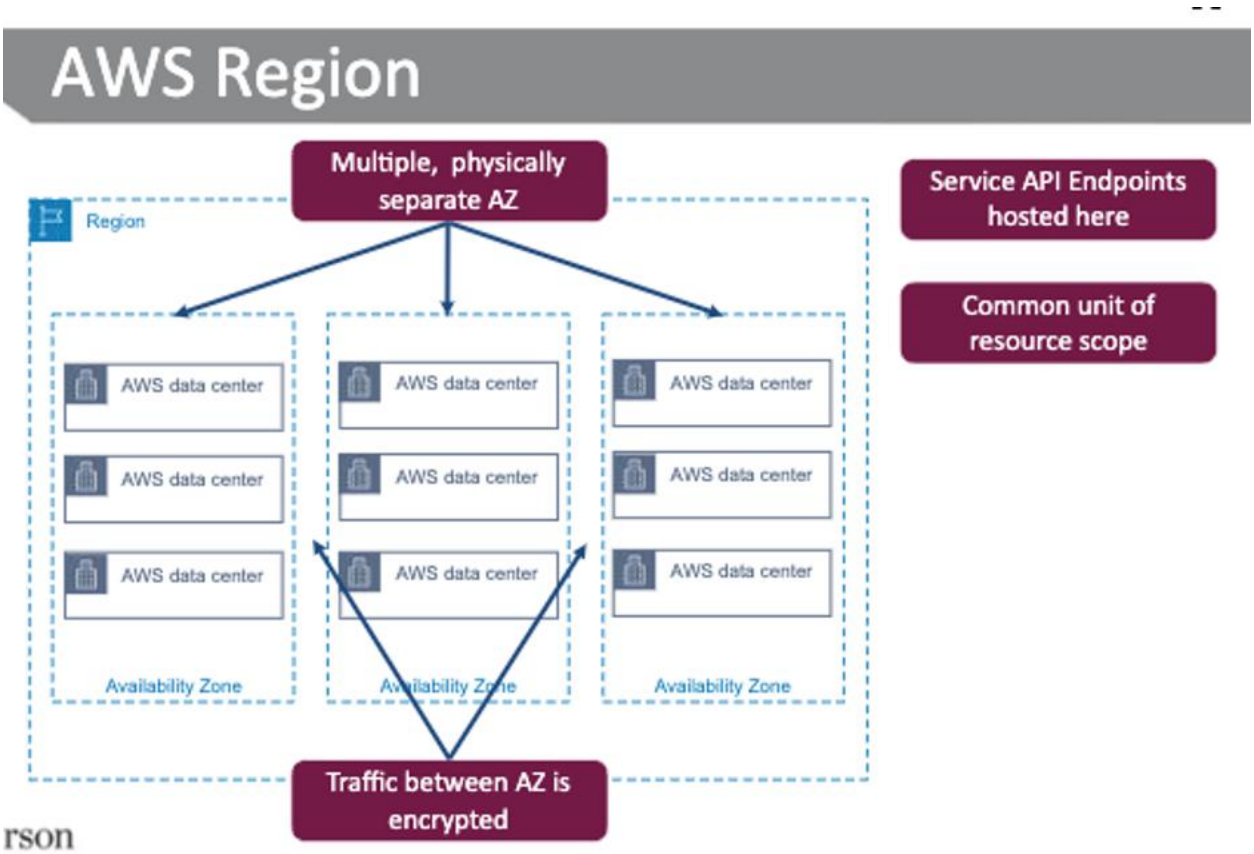
AWS has one of the world's largest cloud infrastructures with:

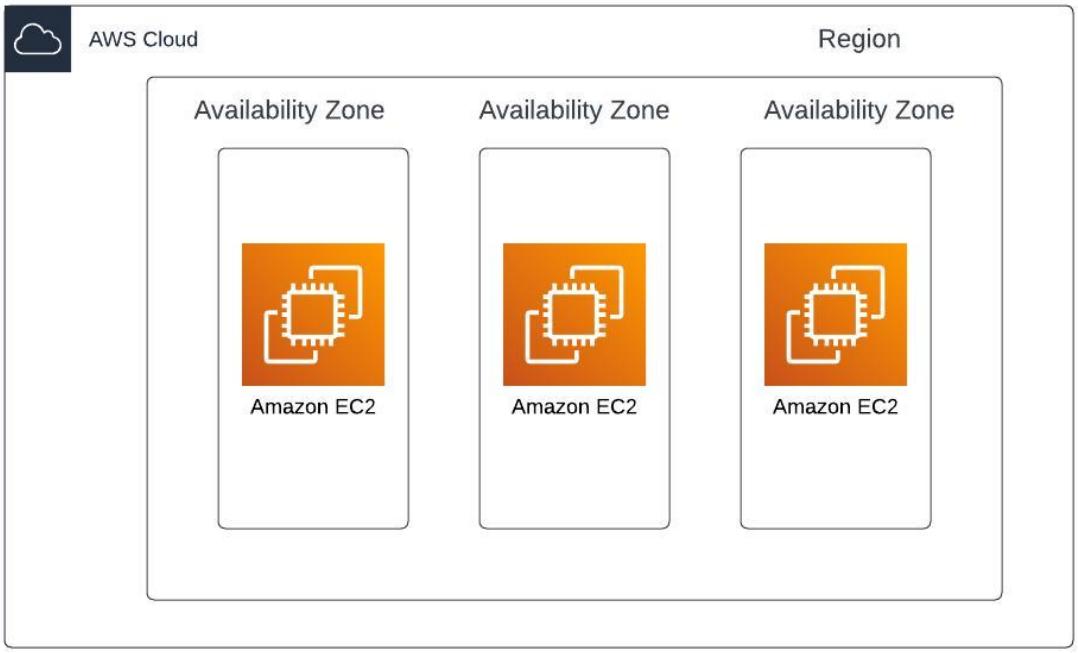
- **Multiple Regions**
- **Availability Zones**
- **Data Centers**
- **Edge Locations**

AWS currently operates many Regions and Availability Zones globally.

AWS Global Infrastructure Diagram







AWS infrastructure is organized as:

Region
↓
Availability Zone
↓
Data Center
↓
Rack
↓
Physical Servers

AWS Regions contain multiple isolated Availability Zones connected using low-latency networking.

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3. Regions

A Region is a geographical location where AWS maintains infrastructure.

Examples

- **Mumbai**
- **Singapore**
- **Tokyo**
- **US East (N. Virginia)**

Why Regions?

- **Low latency**
- **Data compliance**
- **High availability**
- **Better performance**

Diagram

```
World
 ├── India Region
 ├── US Region
 ├── Europe Region
 └── Singapore Region
```

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4. Availability Zones (AZs)

Each AWS Region contains multiple Availability Zones.

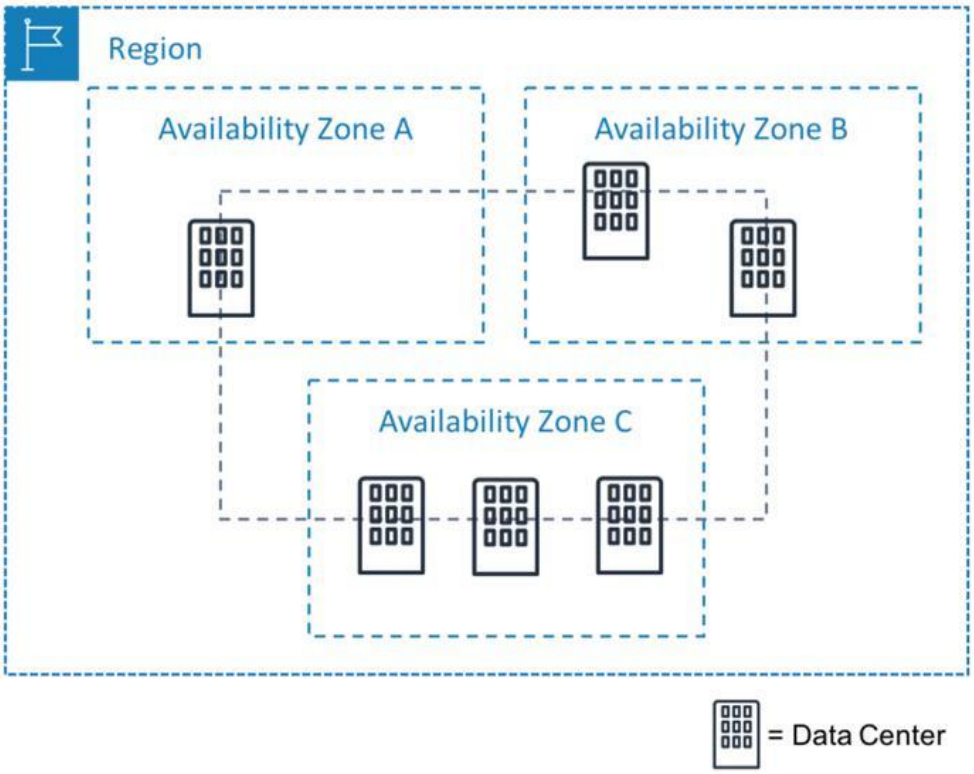
Each AZ contains:

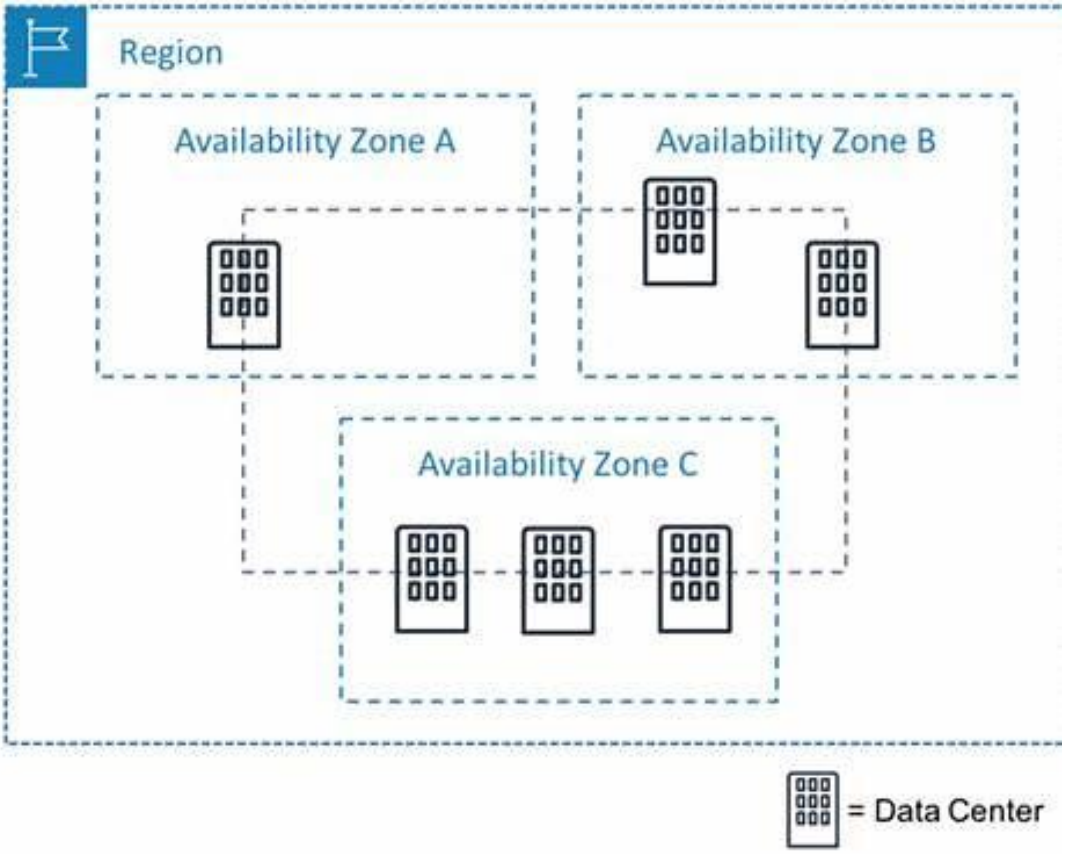
- **One or more Data Centers**
- **Separate power**
- **Separate networking**
- **Separate cooling**

AWS recommends deploying applications across multiple AZs for fault tolerance.

Availability Zone Diagram

Services


Region
Availability Zone A
Availability Zone B
Availability Zone C
= Data Center


Region
Availability Zone A
Availability Zone B
Availability Zone C
= Data Center

Example

Mumbai Region:

- **ap-south-1a**
- **ap-south-1b**
- **ap-south-1c**

Diagram

```
Mumbai Region
  ├── AZ-1
  ├── AZ-2
  └── AZ-3
```

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5. Data Centers, Racks & Physical Servers

Data Center

A secure building that contains:

- **Servers**
- **Storage**
- **Networking devices**
- **Cooling systems**
- **Power systems**

Rack

A rack contains multiple physical servers.

Rack Diagram

```
Data Center
  ↓
 Rack
  ├── Server 1
  ├── Server 2
```

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```
  ├── Server 3
  └── Server 4
```

Physical Server

A physical server is actual hardware containing:

- **CPU**
- **RAM**
- **Storage**
- **Network cards**

Without virtualization:

1 Physical Server = 1 Operating System

With virtualization:

```
1 Physical Server
  ├── VM-1 Web Server
  ├── VM-2 App Server
  └── VM-3 Database Server
```

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6. High Availability (HA)

High Availability means deploying applications across multiple Availability Zones inside the SAME Region.

Goal

Avoid downtime.

HA Diagram

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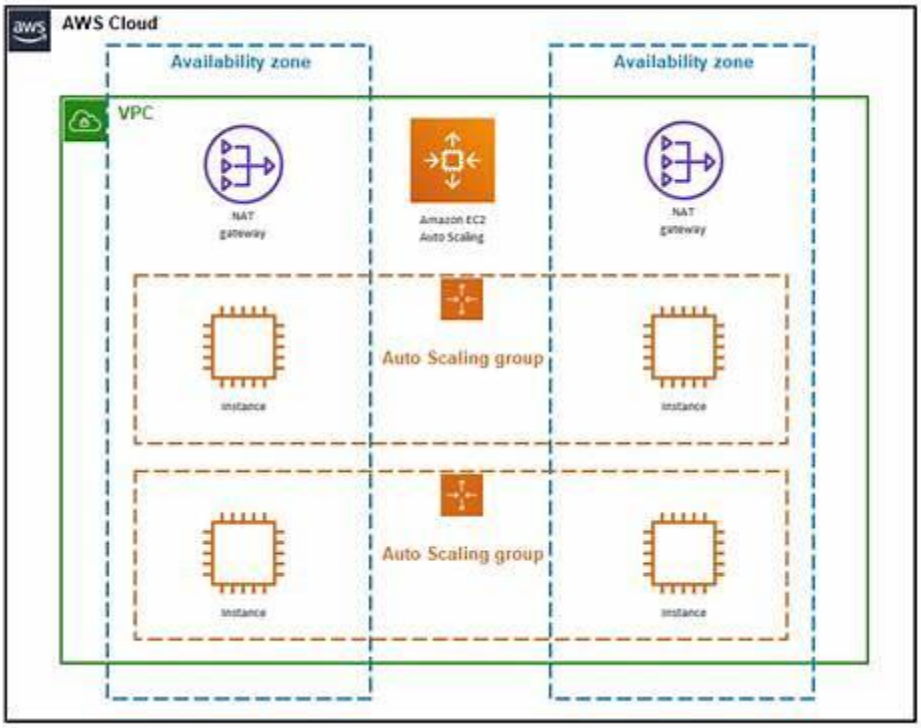

Example

Application deployed in:

- **AZ-1**
- **AZ-2**
- **AZ-3**

If one AZ fails:

- **Application continues from another AZ.**

AWS recommends multi-AZ deployments to improve reliability and fault tolerance.

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7. Disaster Recovery (DR)

Disaster Recovery means deploying applications across MULTIPLE Regions.

Example

- **Primary Region → Mumbai**
- **Backup Region → Singapore**

DR Diagram

Mumbai Region → Primary Site
Singapore Region → Disaster Recovery Site

Difference Between HA & DR

Feature	High Availability	Disaster Recovery
Deployment	Multiple AZs	Multiple Regions
Purpose	Prevent local failure	Recover from regional disaster
Cost	Medium	High

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8. Virtualization

Virtualization allows multiple virtual machines to run on a single physical server.

Benefits

- **Better resource utilization**
- **Reduced hardware cost**
- **Easy scaling**
- **Isolation between applications**

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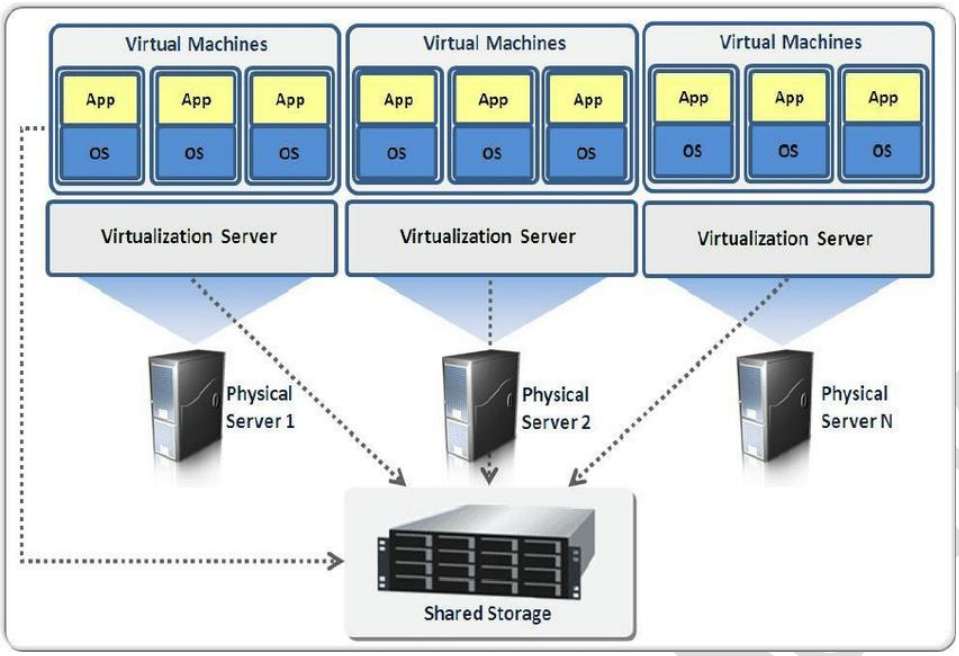
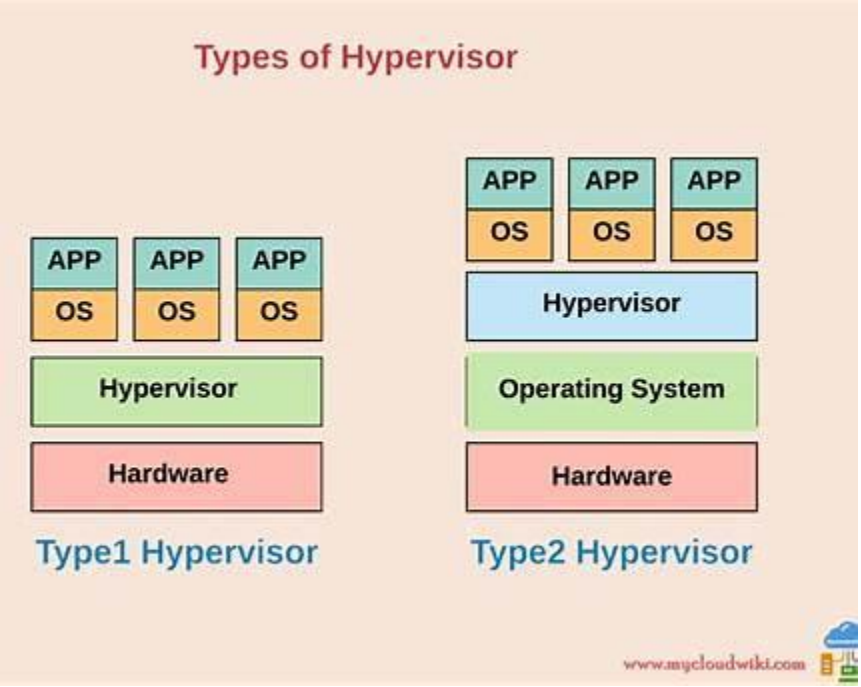
Virtualization Diagram





Virtualization Stack

Physical Hardware

↓

Hypervisor

↓

Virtual Machines

The hypervisor allocates CPU, memory, and storage resources to each virtual machine.

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9. Hypervisors

A Hypervisor is software used to create and manage Virtual Machines.

Common Hypervisors

Provider	Hypervisor
AWS	Xen
Azure	Hyper-V
Google Cloud	KVM
VMware	ESXi

Diagram

Physical Server

↓

Hypervisor

↓

VM-1 VM-2 VM-3

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10. Business Value of Cloud

Companies move to cloud for many advantages.

A. Cost Savings

No need to maintain:

- **Physical servers**
- **Cooling**
- **Data center buildings**

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B. Faster Deployment

Traditional setup:

- **Weeks or months**

Cloud setup:

- **Minutes**

C. Scalability

Servers can increase automatically during traffic spikes.

Diagram

Low Traffic → 2 Servers
High Traffic → 10 Servers

D. Flexibility

Cloud provides:

- **Small servers**
- **Large servers**
- **GPU servers**
- **Storage options**

Important Warning for Students

Cloud is easy to start, but:

- **Unused resources cost money**
- **Poor monitoring increases bills**
- **Proper architecture is important**

Real-Time Interview Question

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Question

Why should applications be deployed across multiple Availability Zones?

Answer

Applications are deployed across multiple Availability Zones to achieve High Availability. If one Availability Zone fails because of power, networking, or hardware issues, the application continues running from another Availability Zone without downtime.

Summary

Today we learned:

- **Methods of connecting to AWS**
- **AWS Global Infrastructure**
- **Regions and Availability Zones**
- **Data Centers and Physical Servers**
- **High Availability and Disaster Recovery**
- **Virtualization and Hypervisors**
- **Business benefits of cloud computing**

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